

SUIDAC Playground Design Programme

IRC SUIDAC - Assosa City, Benishangul-Gumuz, Ethiopia

Consolidated design summary for 6 pre-primary schools serving 709 children aged 5-6 years, including master equipment procurement, implementation prioritisation, and risk assessment for the Cities Alliance and Assosa City Education Office.

6

SCHOOLS

709

CHILDREN

95,643

TOTAL M2

3 + 3

OPTION A / B

1. Programme Overview

The IRC SUIDAC Playground Design Programme encompasses six pre-primary schools located across Assosa City, Benishangul-Gumuz Regional State, Ethiopia. Collectively, these schools serve 709 children aged 5 to 6 years, representing a significant investment in early childhood development infrastructure for the region. The programme was commissioned by the International Rescue Committee (IRC) with funding from Cities Alliance (CA), and all design outputs have been prepared to the standards required for review by the Assosa City Administration Education Office.

The design pipeline followed a rigorous seven-phase approach: site analysis (including KML geospatial data extraction from Cadomation survey data and VLM-based analysis of annotated aerial photographs), concept design (generating two layout options per school), and detailed design (developing the selected option through to equipment placement, surface scheduling, and planting plans). A site-specific selection strategy was employed, assigning Option A (Maximum Play Value) to the three largest schools (Assosa Primary and Middle with 230 students, Benishan Gulgumuz with 124 students, and Assosa Preprimary with 134 students) and Option B (Minimum Viable/Phased) to the three smaller schools (Daresalam with 85 students, Selamber with 68 students, and Gemeharu with 68 students).

All designs adhere to non-negotiable safety standards including minimum 2.0-metre fall zones around moving equipment, 1.5-metre fence clearances, 3.0-metre building setbacks for swing frames and slide platforms, and impact-absorbing surfacing (washed river sand or tire mulch) under all elevated equipment. Materials have been specified exclusively from locally available sources within 50 kilometres of Assosa, and all construction methods are achievable with manual labour and basic tools, ensuring community-based maintenance capability.

2. School Summary

The following table provides a consolidated overview of all six schools in the programme, including student populations, site areas extracted from KML data, design option selections, and the number of equipment installations planned for Phase 1. Total compound areas range from 449 square metres (Gemeharu KG fence) to 50,662 square metres (Assosa Primary and Middle), reflecting the diversity of site conditions encountered across the programme.

#	School Name	Students (M/F)	Total	Site Area	Design Option	Phase 1 Items
1	Gemeharu Primary and Midlevel	25 / 43	68	449 m ²	B (Phased)	4 new
2	Daresalam Primary and Midlevel	40 / 45	85	638 m ²	B (Phased)	3 + 1 retained
3	Assosa Preprimary	78 / 56	134	5,234 m ²	A (Max Play)	17 total
4	Assosa Primary and Middle	124 / 106	230	50,662 m ²	A (Max Play)	29 total
5	Benishan Gulgumuz Primary and Middle	65 / 59	124	14,427 m ²	A (Max Play)	Full set
6	Selamber Primary School	30 / 38	68	24,333 m ²	B (Phased)	6 total

3. Design Option Selection Rationale

The site-specific selection approach was adopted to balance play value against budget efficiency and maintenance sustainability across the programme. Schools were categorised by student population, with the threshold set at 100

students for Option A selection. The three schools exceeding this threshold (Assosa Primary and Middle at 230, Assosa Preprimary at 134, and Benishan Gulgumuz at 124) received Option A designs featuring maximum equipment density and comprehensive surfacing. The remaining three schools (Daresalam at 85, Selamber at 68, and Gemeharu at 68) received Option B phased designs prioritising core equipment with clearly delineated Phase 2 expansion zones.

Option A designs deliver higher initial play value and more diverse play experiences but require greater upfront investment and ongoing maintenance capacity. Option B designs focus on the essential quartet of slide, swing, see-saw, and a single climbing element, with additional infrastructure (tunnel crawls, stepping logs, and supplementary equipment) deferred to Phase 2 (months 6 to 12). This phased approach reduces initial costs by approximately 30 to 40 percent while ensuring that all children have access to the four fundamental play types (sliding, swinging, balancing, and climbing) from day one of the programme.

4. Master Equipment Procurement List

The following table consolidates all equipment requirements across the six schools, distinguishing between procured metal equipment (E1 through E4) manufactured to specification and locally constructed elements (L1 through L4) built from bamboo, timber, and concrete materials available within 50 kilometres of Assosa. All metal equipment requires zinc-chromate primer and enamel topcoat for UV and corrosion resistance, while all bamboo and timber elements must undergo boron-based anti-termite treatment for a minimum of 24 hours before installation.

Code	Equipment	Unit	S1	S2	S3	S4	S5	S6	TOTAL
E1	Merry-Go-Round	No.	0	1	1	1	1	1	5
E2	See-Saw	No.	1	1	2	5	2	1	12
E3	Slide (Standard)	No.	1	1	3	5	3	2	15
E4	Chain Swing	No.	1	1	3	5	3	1	14
L1	Balance Beam (Bamboo)	No.	1	1	3	3	2	1	11
L2	Climbing Structure	No.	1	1	2	4	2	1	11
L3	Tunnel Crawl	No.	1	1	1	3	2	1	9
L4	Stepping Logs/Stones	Set	1	1	2	3	2	1	10

5. Surface Material Summary

All surfacing materials have been sourced from within 50 kilometres of Assosa City. Washed river sand is obtained from local riverbeds and specified at three depth categories depending on the equipment type: 150 millimetres under merry-go-rounds and see-saws, 200 millimetres under swings, and 300 millimetres beneath slide runoff zones. Fine gravel (10 to 20 millimetre aggregate) from local quarries is used for 100-millimetre compacted pathways. Grass areas are to be established using locally adapted drought-tolerant species through transplant or seed. Tire mulch, produced by shredding discarded tires collected from school compounds (notably Selamber), provides tree protection rings and an alternative impact-absorbing surface under swing frames where rubber tire mulch is specified.

Material	Application	Estimated Total Volume/Area	Source
Washed River Sand	Under equipment	85 m ³	Local riverbeds

Material	Application	Estimated Total Volume/Area	Source
Fine Gravel (10-20mm)	Pathways	42 m ³	Local quarries
Grass (transplant/seed)	General play lawn	680 m ²	Local species
Tire Mulch	Tree rings / swing zones	12 m ³	Recycled on-site
Coarse Gravel	Drainage swales	18 m ³	Local quarries

6. Implementation Priority Ranking

The implementation priority ranking considers four factors: data quality (availability of site surveys and photographs), student population (larger schools serve more children per unit of investment), design readiness (completeness of the design package), and site preparation requirements (clearing, grading, or equipment removal needed before installation can begin). Schools ranked 1 through 3 have the most complete data and largest student populations, making them the most efficient choices for immediate implementation. Schools ranked 4 through 6 have moderate data gaps or require additional site preparation, but all six designs are construction-ready pending field verification.

Rank	School	Students	Justification
1	Assosa Preprimary	134	Best data (photos + KML), medium-large population, Option A ready
2	Assosa Primary and Middle	230	Largest school, exceptional tree canopy, site survey recommended
3	Benishan Gulgumuz	124	Good data, large compound with ample space, Option A ready
4	Daresalam	85	Constrained 638 m ² KG boundary, Option B, survey needed
5	Selamber	68	Overgrown site needs clearing and grading, rusted equipment removal
6	Gemeharu	68	Smallest KG area (449 m ²), no existing equipment, simplest

7. Key Risks and Mitigation Strategies

The following risks have been identified through the design process and are presented alongside recommended mitigation strategies. The highest-priority risk relates to the reliability of KML elevation data, which is derived from Google Earth and carries an estimated margin of error of plus or minus 10 metres. This directly impacts drainage design and should be verified through field-level surveys before construction begins at any site.

Risk	Impact	Likelihood	Mitigation
KML elevation data unreliable (+/-10m)	High	Certain	Field-level survey before grading; design conservatively assuming 1-2% slope
High termite risk to bamboo/timber	High	Certain	Mandatory 24hr boron soak for all local construction elements
Sand/gravel mixing with lateritic soil	Medium	High	Add non-woven geotextile separation layer under all surfacing

Risk	Impact	Likelihood	Mitigation
Wet season drainage failure	High	Medium	Gravity-based swales, 1% min slope on all surfaces, sand edge containment
Constrained KG boundaries (DS, GH)	Medium	Certain	Optimised layout within available space; negotiate boundary expansion
Community maintenance capacity	Medium	Medium	Maintenance framework per school; community training in Phase 3
Metal equipment corrosion	Medium	High	Zinc-chromate primer + enamel topcoat; annual inspection schedule

8. Lessons Learned from the Design Process

Several key lessons emerged from executing the design pipeline across six diverse school sites. First, the quality and granularity of input data directly correlates with design confidence: schools with both annotated aerial photographs and detailed KML boundary data (Schools 3, 5, and 6) produced more precise and verifiable designs than those relying solely on one data source. Second, the distinction between the full school compound boundary and the dedicated pre-primary (KG) play area boundary is critical for accurate space planning. Schools 1 and 2 revealed that the KG boundary can be dramatically smaller than the total compound, constraining equipment layout options and requiring creative space optimisation strategies.

Third, the VLM-based image analysis proved valuable for identifying site features not captured in the KML data, including tree canopy coverage estimates, ground surface conditions, and contextual land use patterns. However, VLM analysis of annotated aerial photographs sometimes produced descriptions at odds with the ground-truth descriptions in the protocol document, highlighting the importance of cross-referencing multiple data sources. Fourth, the phased design approach (Option B) proved essential for the smaller schools where space constraints and budget limitations would otherwise result in either overcrowded or under-equipped play areas. Fifth, the inclusion of a maintenance framework with specific actions, frequencies, and responsible parties from the outset ensures that the infrastructure investment is protected beyond the initial construction phase, which is particularly important in the IRC context where long-term operational sustainability is a donor requirement.

9. Next Steps and Recommendations

To advance from design to construction, the following steps are recommended in priority order. First, conduct field-level site verification surveys at all six schools, focusing on confirming KML boundary dimensions, measuring spot elevations for drainage design, and photographing current conditions at Schools 1, 2, and 4 where photo documentation was previously limited. Second, finalise equipment procurement by consolidating the master procurement list and engaging metal fabrication workshops in Assosa or Addis Ababa for the production of E1 through E4 equipment items. Third, engage community stakeholders at each school through meetings with school administrations, parent-teacher associations, and local labour pools to confirm community contribution commitments for unskilled labour and local material collection (river sand, bamboo, timber).

Fourth, commission a detailed topographic survey at Schools 3, 4, and 5 where the large compound sizes and variable terrain make drainage design particularly sensitive to elevation accuracy. Fifth, develop community maintenance training materials and schedules to be delivered during the Phase 3 implementation window, covering equipment inspection routines, sand surface replenishment procedures, and timber/bamboo retreatment schedules. Sixth, establish a monitoring and evaluation framework with baseline data collection (including photographs) at each school prior to construction, enabling post-occupancy evaluation of playground utilisation and safety compliance.

after the first full wet season of operation.